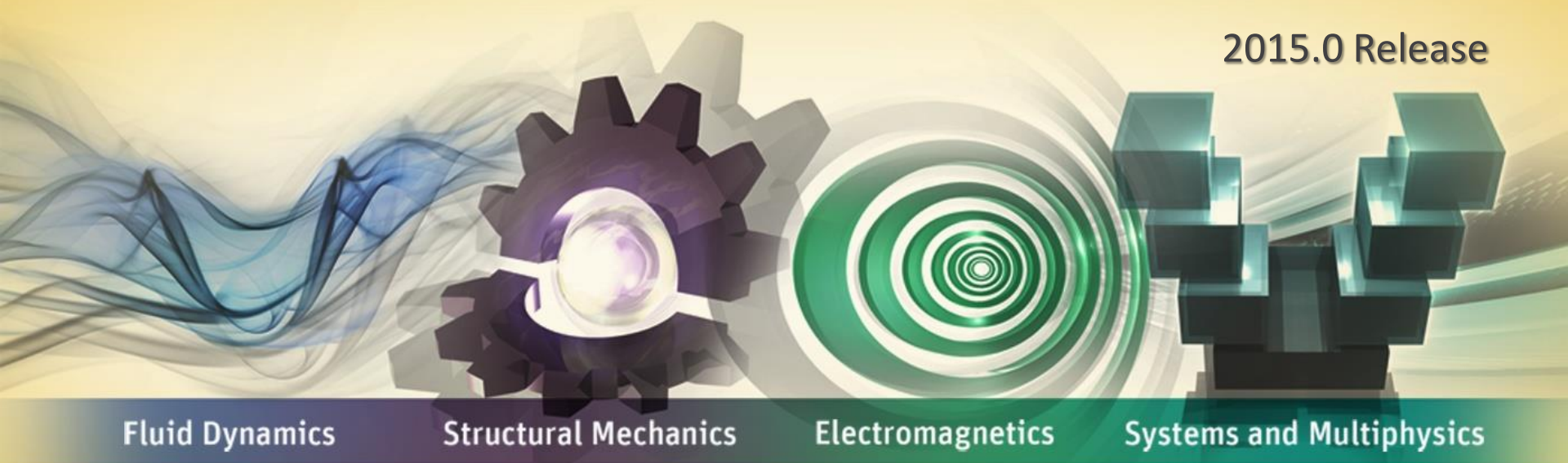


Workshop 4-1: Radiation Boundaries

2015.0 Release

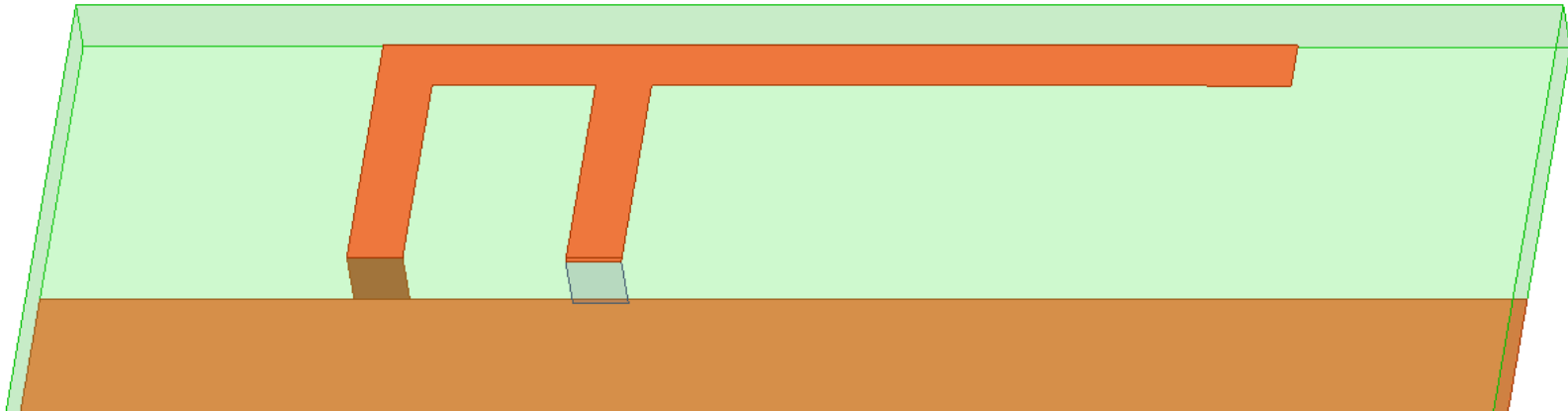


ANSYS HFSS for Antenna Design

Example: Radiation Boundaries

- **Planer Inverted-F Antenna**

- This example is intended to show you how to apply a Radiation Boundary and how its size can impact the solution.



- **Overview**

- Antenna simulated using an ABC (Absorbing Boundary Condition)
 - Parametric Analysis
- Antenna simulated using an PML (Perfectly Match Layer)
 - Three Simulations $\lambda/20$, $\lambda/10$ and $\lambda/4$
- Antenna Simulated using FE-BI
 - Parametric Analysis

- **Launching ANSYS Electronics Desktop 2015**

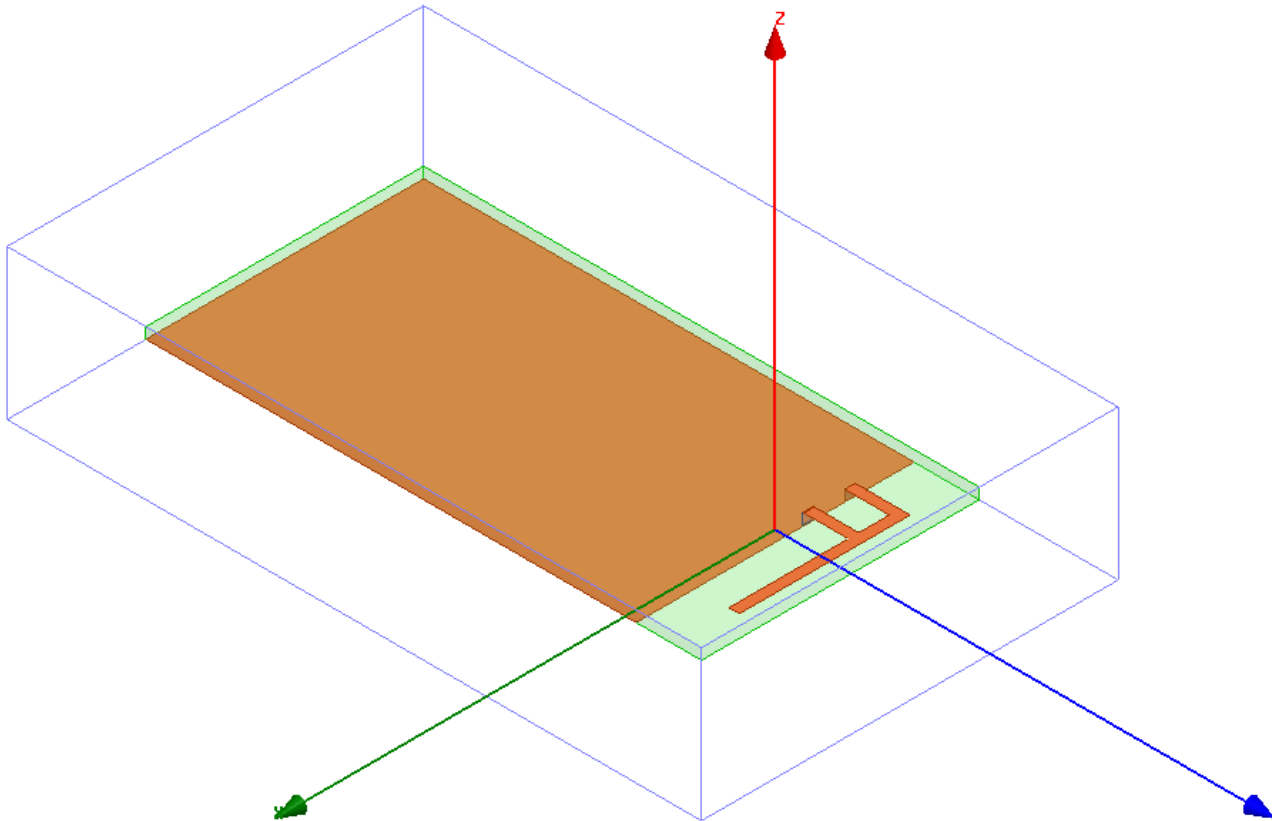
- To access ANSYS Electronics Desktop, click the Microsoft Start button,
 - Select *Programs > ANSYS Electromagnetics > ANSYS Electromagnetics Suite 16.0*. Select *ANSYS Electronics Desktop 2015*

- **Setting Tool Options**

- **Note:** In order to follow the steps outlined in this example, verify that the following tool options are set:
- Select the menu item *Tools > Options > HFSS Options...*
 - Click the **General** tab
 - Use Wizards for data input when creating new boundaries: ☒ **Checked**
 - Duplicate boundaries/mesh operations with geometry: ☒ **Checked**
 - Click the **OK** button
- Select the menu item *Tools > Options > 3D Modeler Options....*
 - Click the **Operation** tab
 - Select last command on object select: ☒ **Checked**
 - Click the **Display** tab
 - Set default transparency to **0.7**
 - Click the **Drawing** tab
 - Edit properties of new primitives: ☒ **Checked**
 - Click the **OK** button

- **Opening a Project**

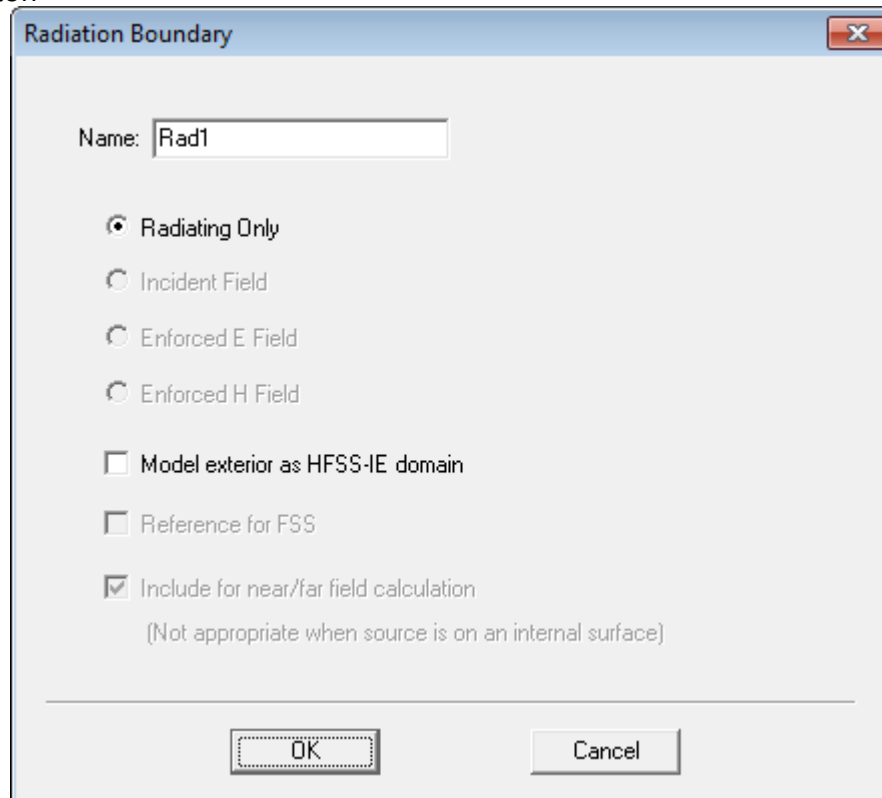
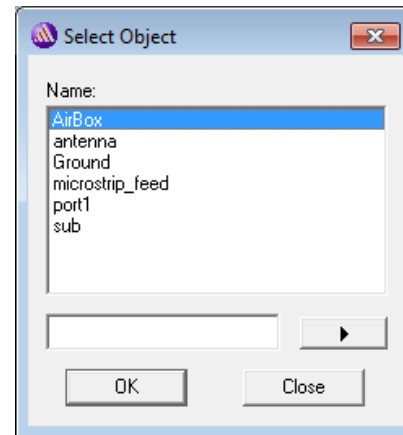
- In the ANSYS Electronics Desktop, select the menu item **File > Open**
 - Browse to the folder containing the file **PIFA_Training.aedt** and select **Open**
 - Select the menu item **File > Save As**.
 - From the **Save As** window, type the Filename: **PIFA_W_ABC.aedt**



Radiation Boundary

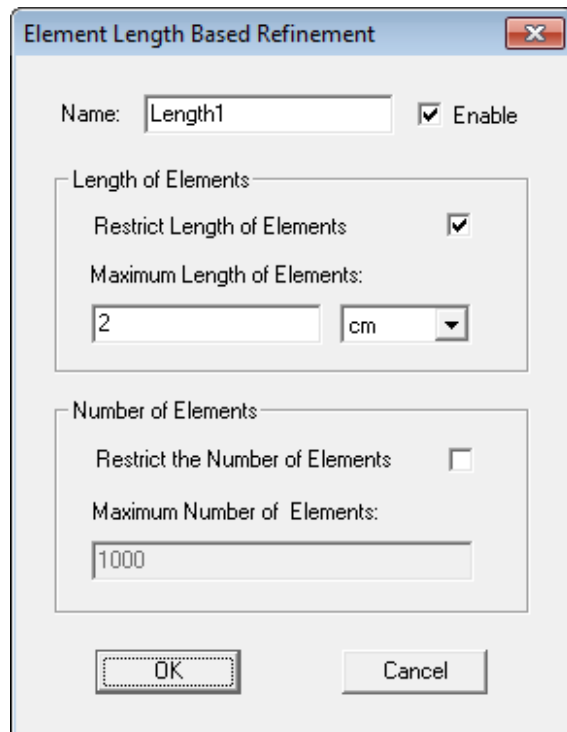
• Applying Radiation Boundary

- Select the menu item **Edit > Select > Objects**
- Select the menu item **Edit > Select > By Name...**
 - Select **AirBox** and click **OK**
- Select the menu item **Edit > Select > All Object Faces**
- Select the menu item **HFSS > Boundaries > Assign > Radiation...**
 - Click the **OK** button



- **Applying Mesh Operation on Radiation Boundary**

- Select the menu item **Edit > Select > Objects**
- Select the menu item **Edit > Select > By Name...**
 - Select **AirBox** and click **OK**
- Select the menu item **Edit > Select > All Object Faces**
- Select the menu item **HFSS > Mesh Operations > Assign > On Selection > Length Based**
 - Maximum Length of Elements: **2cm**
- Click the **OK** button



Note: To improve far field results for any distance, the radiation boundary needs to be seeded with $\lambda/6$ mesh operation at the highest frequency.

Solution frequency is 2.5GHz, $\lambda = 12\text{cm}$, $\lambda/6 = 2\text{cm}$

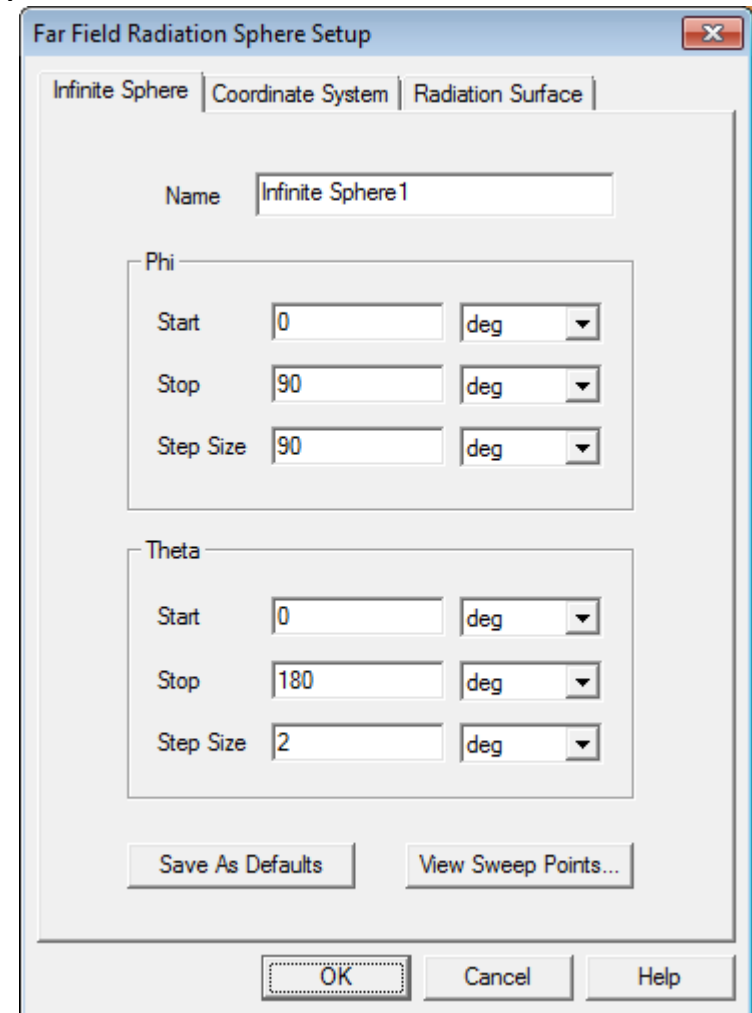
Therefore we set the mesh operation to 2cm

- **Create a Radiation Setup**

- Select the menu item **HFSS > Radiation > Insert Far Field Setup > Infinite Sphere...**
 - In the **Infinite Sphere** Tab
 - Name: **Infinite Sphere 1**
 - Phi:
 - Start: **0 deg**
 - Stop: **90 deg**
 - Step Size: **90 deg**
 - Theta:
 - Start: **0 deg**
 - Stop: **180 deg**
 - Step Size: **2 deg**
 - Click the **OK** button

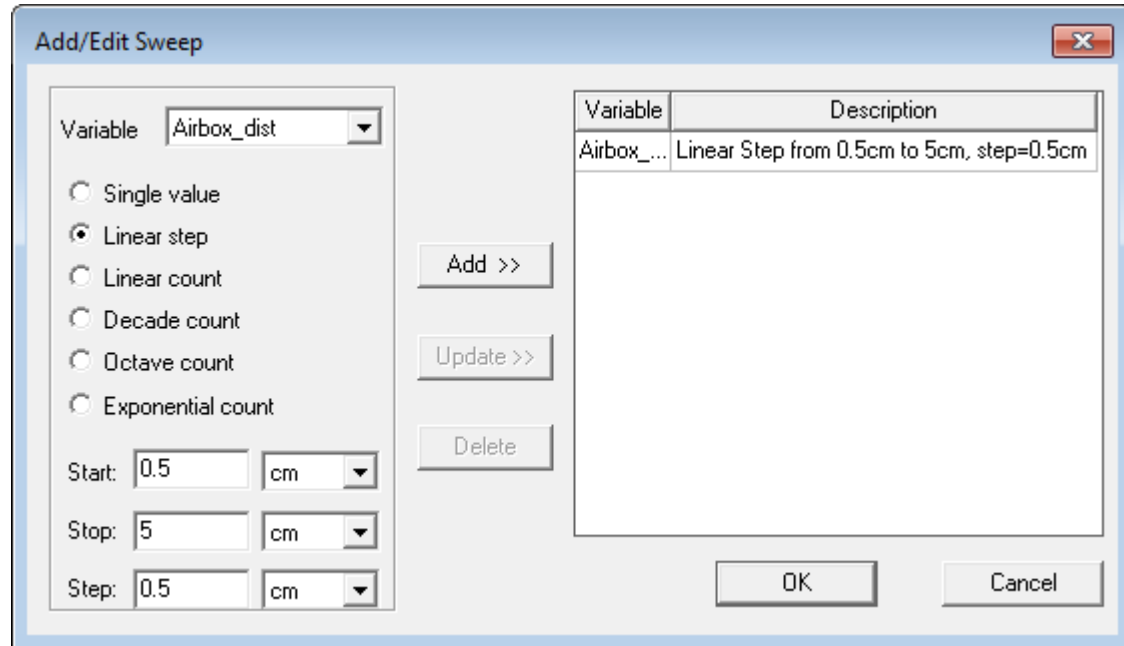
- **Model Validation**

- Select the menu item **HFSS > Validation Check**
- Click the **Close** button



• Creating a Parametric Sweep

- Select the menu item **HFSS > Optimetrics Analysis > Add Parametric...**
 - In the **Sweep Definitions** tab.
 - Click the **Add...** button
 - Variable: **AirBox_dist**
 - Select the radio button: **Linear step**
 - Start: **0.5 cm**
 - Stop: **5 cm**
 - Step: **0.5 cm**
 - Click the **Add>>** button
 - Click the **OK** button
 - Click the **Options** tab.
 - Save Fields and Mesh: ☒ **Checked**
 - Click the **OK** button



• Save Project

- Select the menu item **File > Save**

• Analyze the Project

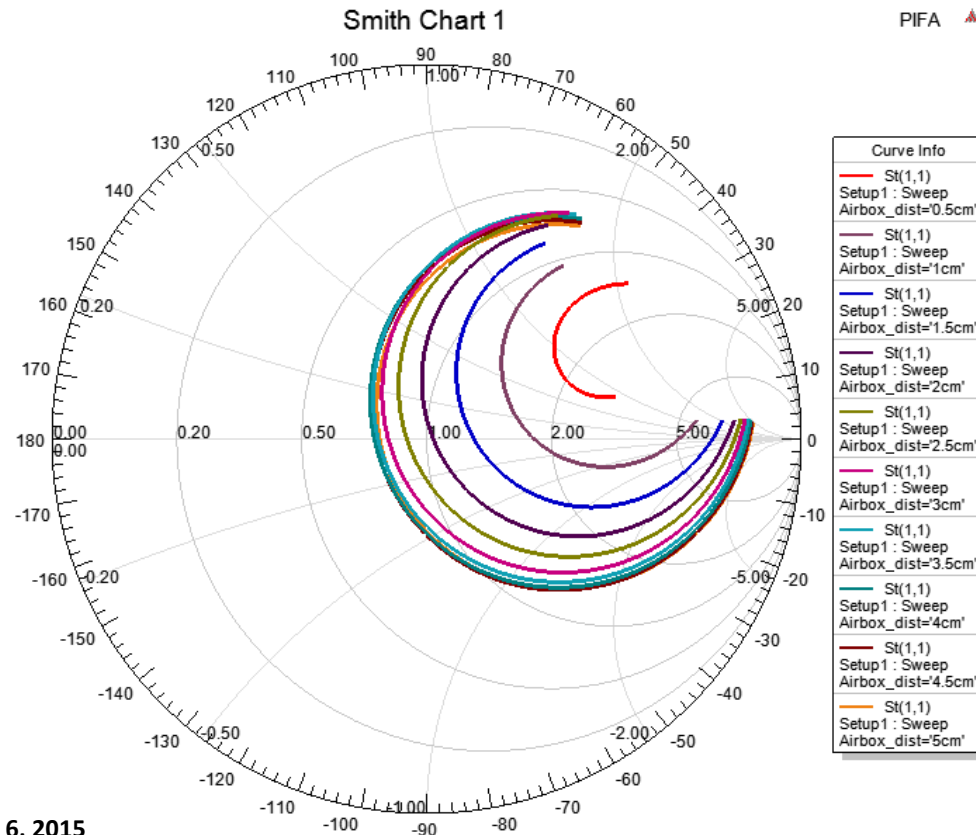
- Select the menu item **HFSS > Analyze All**

Note: Saving Fields and Mesh will save all the field data for all Discrete and Last Adaptive solutions. To reduce the disk space requirements we can choose to save only the far field data, and not the field data inside of the entire volume.

Create Smith Chart

- Select the menu item **HFSS > Results > Create Terminal Solution Data Report > Smith Chart**
 - Solution: **Setup1: Sweep**
 - In the **Trace** tab:
 - Category: **Terminal S Parameter**
 - Quantity: **St(1,1)**
 - Function: **<none>**
 - Click the **New Report** button
 - Click the **Close** button

Note: The S-parameter stabilizes at $\lambda/4$, from the radiating element. We recommended that you size the air volume $\lambda/4$ at the lowest frequency. This example has a lower sweep point of 2 GHz and a $\lambda/4$ of 4cm.

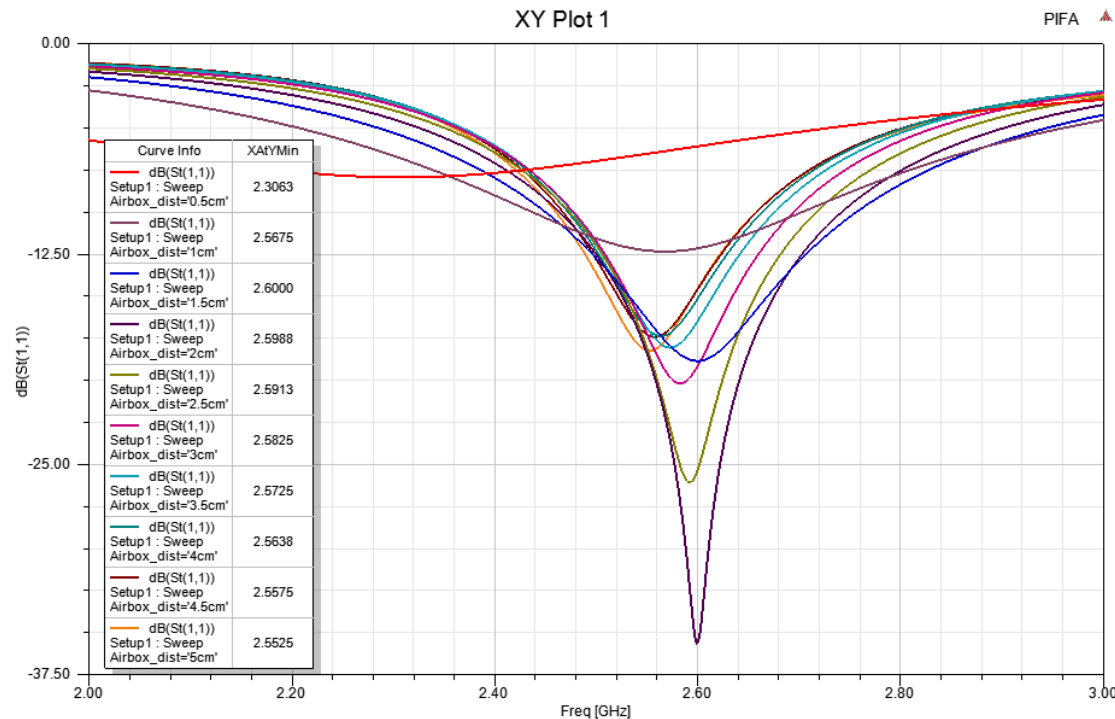
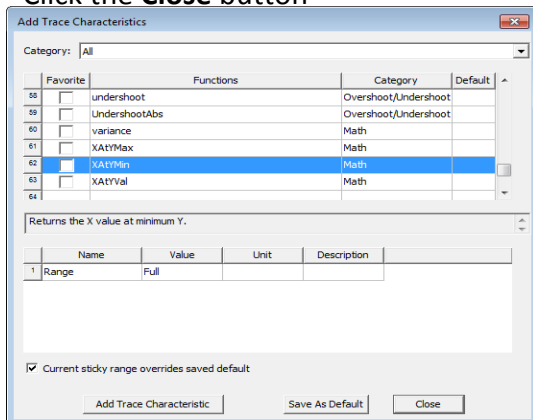


• Create Rectangular Plot

- Select the menu item **HFSS > Results > Create Terminal Solution Data Report > Rectangular Plot**
 - Solution: **Setup1: Sweep**
 - Domain: **Sweep**
 - In the **Trace** tab:
 - Category: **Terminal S Parameter**
 - Quantity: **St(1,1)**
 - Function: **dB**
 - Click the **New Report** button
 - Click the **Close** button

• Trace characteristics

- Right-click in XY Plot1 (in Report window)
- Select **Trace Characteristics > All...**
 - Scroll-down to the bottom and select **XatYMin**
 - Click the **Add Trace Characteristic** button
 - Click the **Close** button



Note: The resonant frequency stabilizes at 2.55 GHz when the radiation boundary is greater than 4 cm away from the antenna. This is the distance where the boundary condition is not loading the antenna but absorbing the radiated energy

• Create Far Field Radiation Pattern

- Select the menu item **HFSS > Results > Create Far Fields Report > Rectangular Plot**

- Solution: **Setup1: LastAdaptive**

- Geometry: **Infinite Sphere1**

- In the **Trace** tab:

- Category: **Realized Gain**
- Quantity: **RealizedGainTotal**
- Function: **dB**

- Click the **Families** Tab

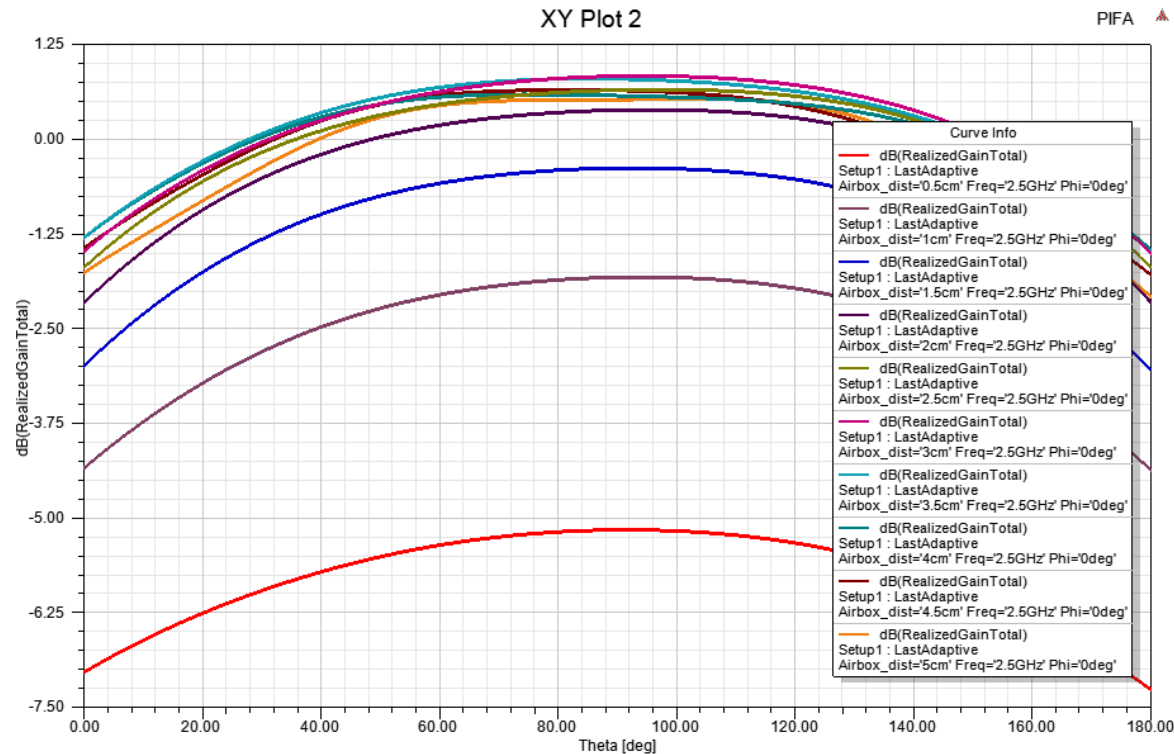
- Click the **Edit** button  associated with **Phi** variable
 - Click **0deg** in the pop-up window
 - Close the pop-up window by clicking the **X** button

- Freq: **2.5 GHz**

- Airbox_dist: **All**

- Click the **New Report** button

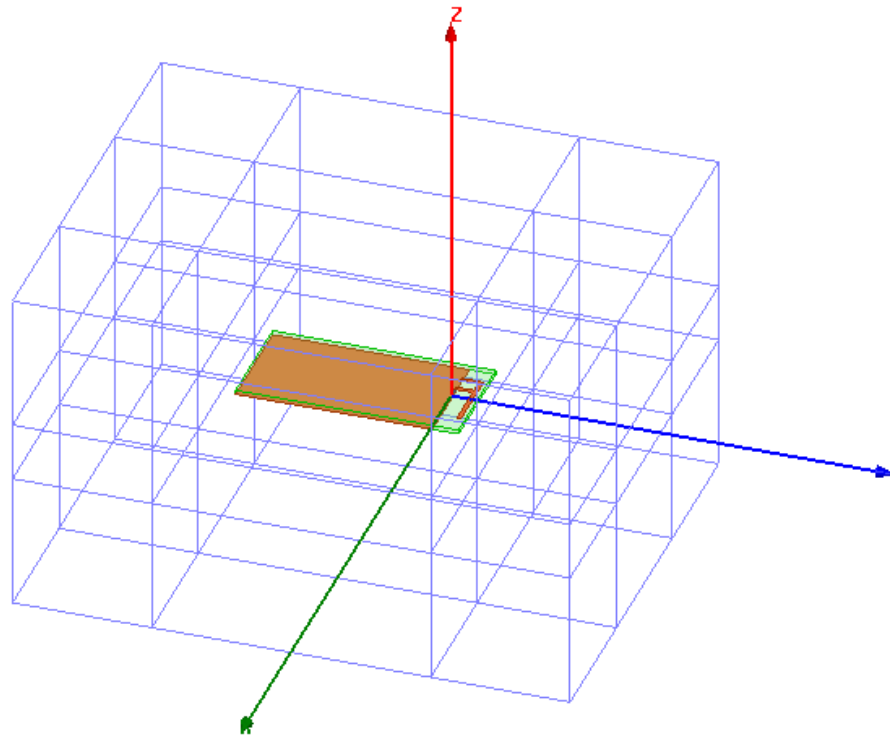
- Click the **Close** button



Note: The peak gain value converges toward .5 dB as the air volume increases beyond $\lambda/4$

Antenna with PML

- Antenna simulated using an PML (Perfectly Match Layer)
 - Three Simulations $\lambda/20$, $\lambda/10$ and $\lambda/4$
- This example will demonstrate the use of the PML boundary condition
- Compare Return Loss for different distances to the PML
- Compare Radiation Patterns for different distances to the PML



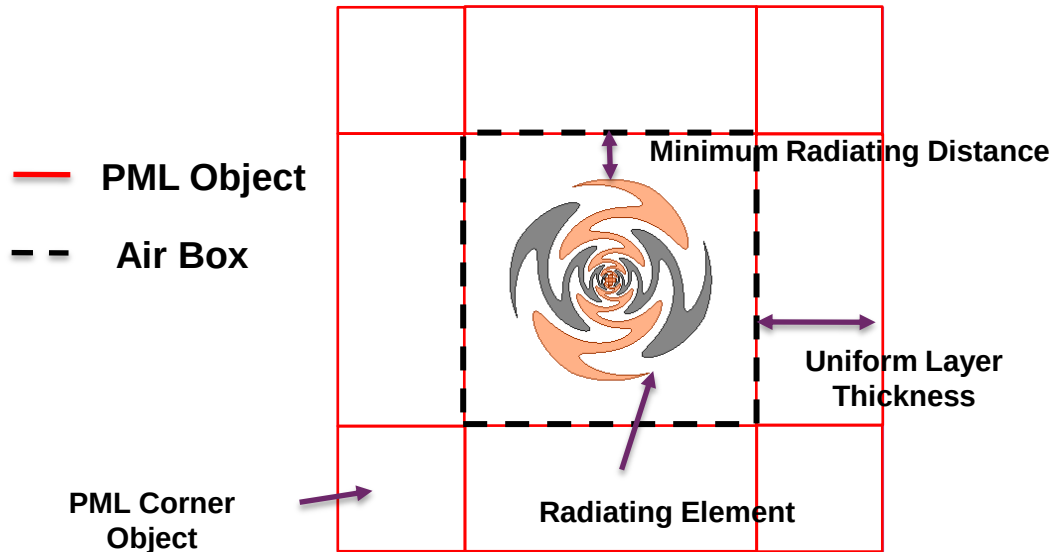
PML Setup Wizard

• PML Setup Wizard

- Helps create objects with appropriate material properties

• PML Inputs

- Uniform Layer Thickness
 - Thickness of PML Object Created (recommended $> \lambda/3$)
- Minimum Frequency
 - Minimum frequency that PML will be absorbing
- Minimum Radiating Distance
 - Distance from radiating object to PML Object (recommended $> \lambda/8$)



1

PML Setup Wizard : Cover Objects

☒ Create PML Cover Objects On Selected Faces

Uniform Layer Thickness: 10 cm

Use Default Formula

☒ Create joining corner and edge objects

☐ Use Selected Object, 'AirBox', as PML Cover
(Use this option to cover non-planar faces)

Corresponding Base Object: AirBox

Uniform Layer Thickness: 0.71916 cm

Orientation in Local Coordinates: ☐ X Axis ☐ Y Axis ☐ Z Axis

☐ Do Not Create New Cover Objects

Base Face Radiation Properties

☒ Radiating Only

☐ Incident Field

< Back Next > Cancel

2

PML Setup Wizard : Material Parameters

☒ PML Objects Accept Free Radiation

Min Frequency: 1 GHz

☐ PML Objects Continue Guided Waves

Propagation Constant at Min Frequency: 20 1/m

Minimum Radiating Distance: 3.75 cm

Use Default Formula

< Back Next > Cancel

• Opening a Project

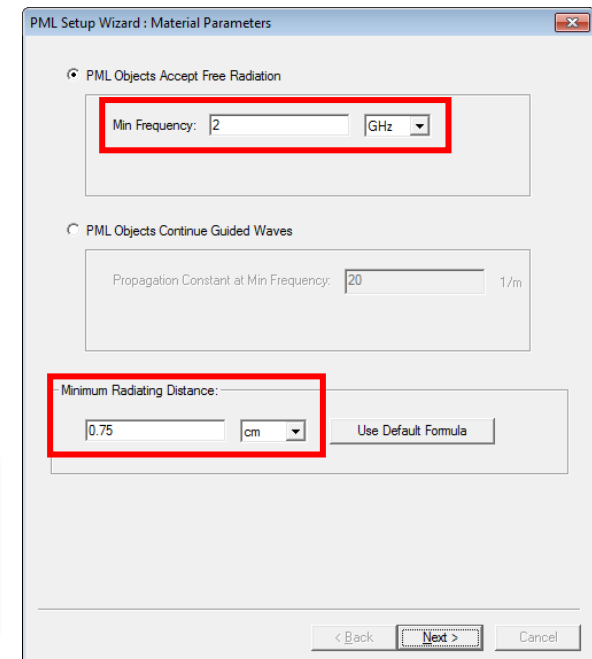
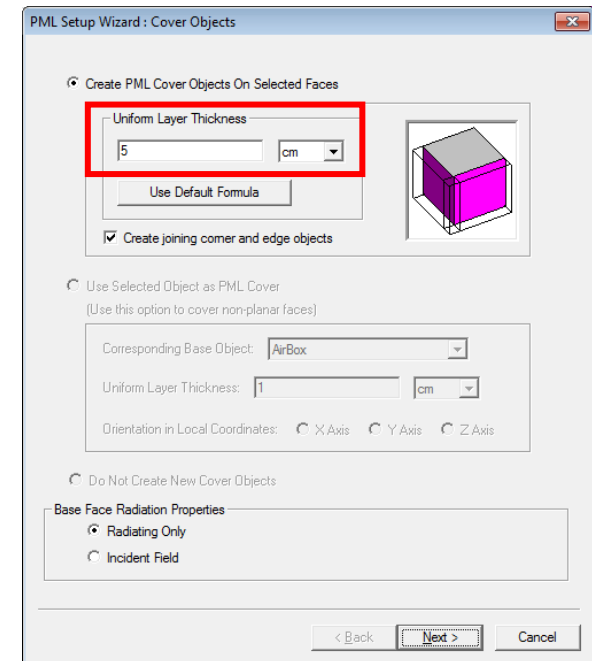
- From the Electronics Desktop, click the  On the Standard toolbar, or select the menu item **File > Open**
- Select the project **PIFA_W_PML.aedt**
- Select Design **"1_PIFA Lambda_by_20"**

• Applying PML Boundary

- Select the menu item **Edit > Select > Objects**
- Select the menu item **Edit > Select > By Name...**
 - Select **AirBox** and click **OK**
- Select the menu item **Edit > Select > All Object Faces**
- Select the menu item **HFSS > Boundaries > PML Setup Wizard...**
 - Select **Create PML Cover objects on Selected Faces** radio button
 - Uniform Layer Thickness: **5 cm**
 - Click the **Next** button
 - Select **PML Objects Accept Free Radiation** radio button
 - Min Frequency: **2 GHz**
 - Minimum Radiating Distance: **0.75 cm**
 - Click the **Next** button
 - Click the **Finish** button

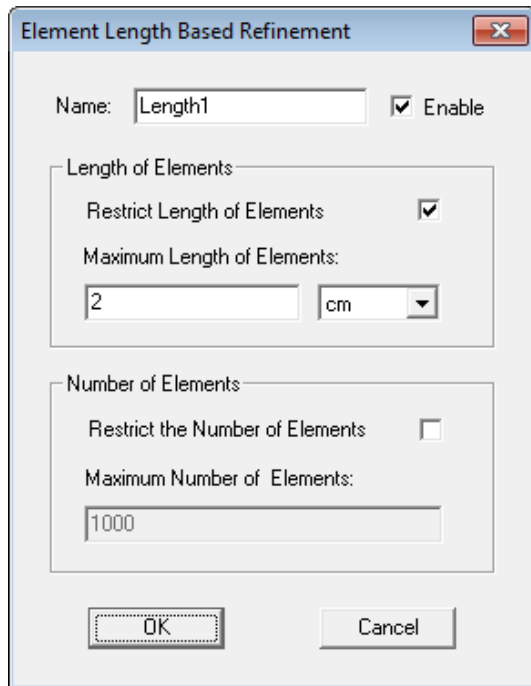
The PML wizard will create a uniform layer thickness, use $\lambda/3$ of lowest frequency $\lambda=15\text{cm}$ @ 2GHz, therefore use a value of 5cm

Distance from radiating object to PML Object ($\lambda/20$) = 0.75cm



- **Applying Mesh Operation on Radiation Boundary**

- Select the menu item **Edit > Select > Objects**
- Select the menu item **Edit > Select > By Name...**
 - Select **AirBox** and click **OK**
- Select the menu item **Edit > Select > All Object Faces**
- Select the menu item **HFSS > Mesh Operations > Assign > On Selection > Length Based...**
 - Maximum Length of Elements: **2cm**
- Click the **OK** button



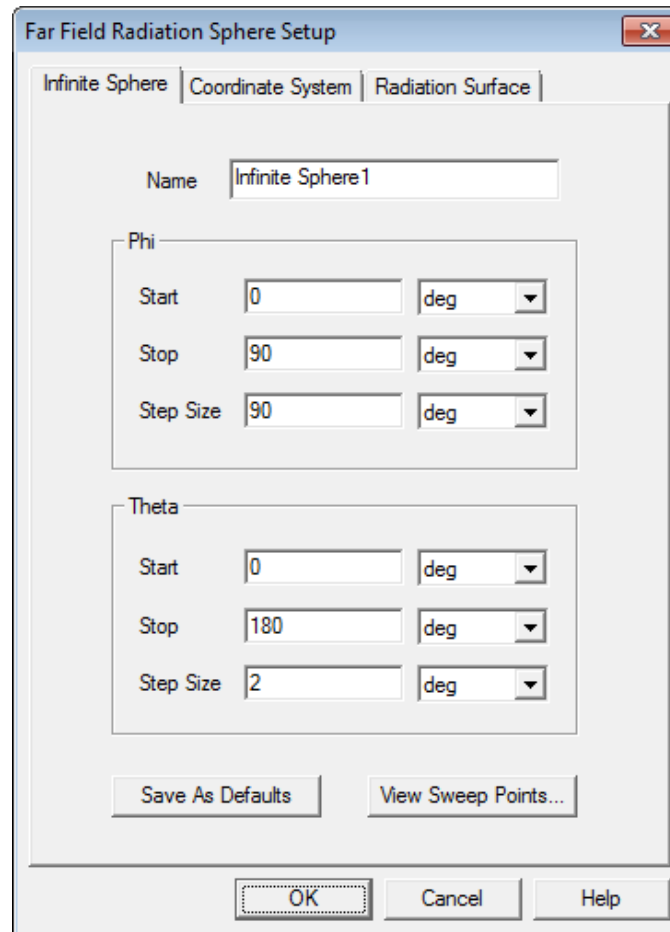
Note: To improve far field results for any distance, the radiation boundary needs to be seeded with $\lambda/6$ mesh operation at the highest frequency.

Solution frequency is 2.5GHz, $\lambda = 12\text{cm}$, $\lambda/6 = 2\text{cm}$

Therefore we set the mesh operation to 2cm

- **Create a Radiation Setup**

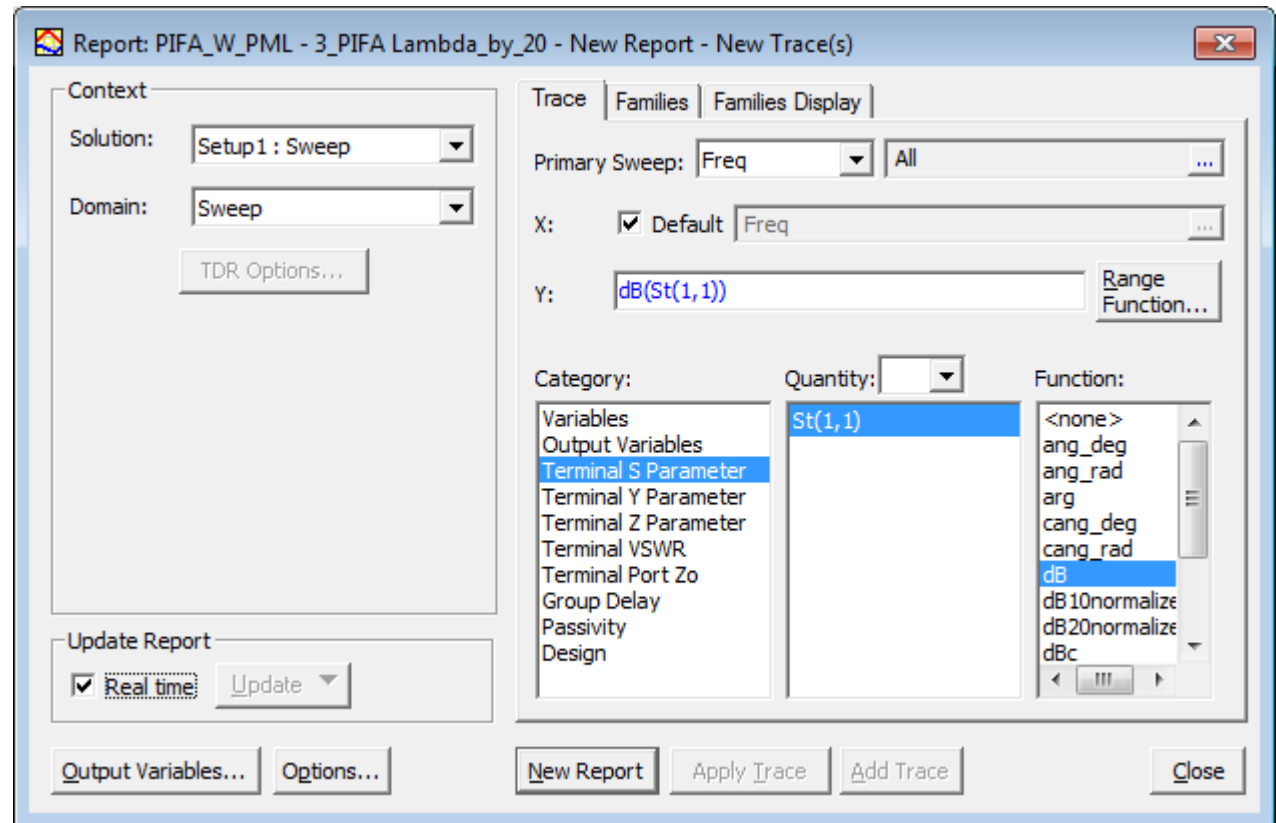
- Select the menu item **HFSS > Radiation > Insert Far Field Setup > Infinite Sphere...**
 - In the **Infinite Sphere** Tab
 - Name: **Infinite Sphere1**
 - Phi:
 - Start: **0 deg**
 - Stop: **90 deg**
 - Step Size: **90 deg**
 - Theta:
 - Start: **0 deg**
 - Stop: **180 deg**
 - Step Size: **2 deg**
 - Click the **OK** button



Rectangular Plot

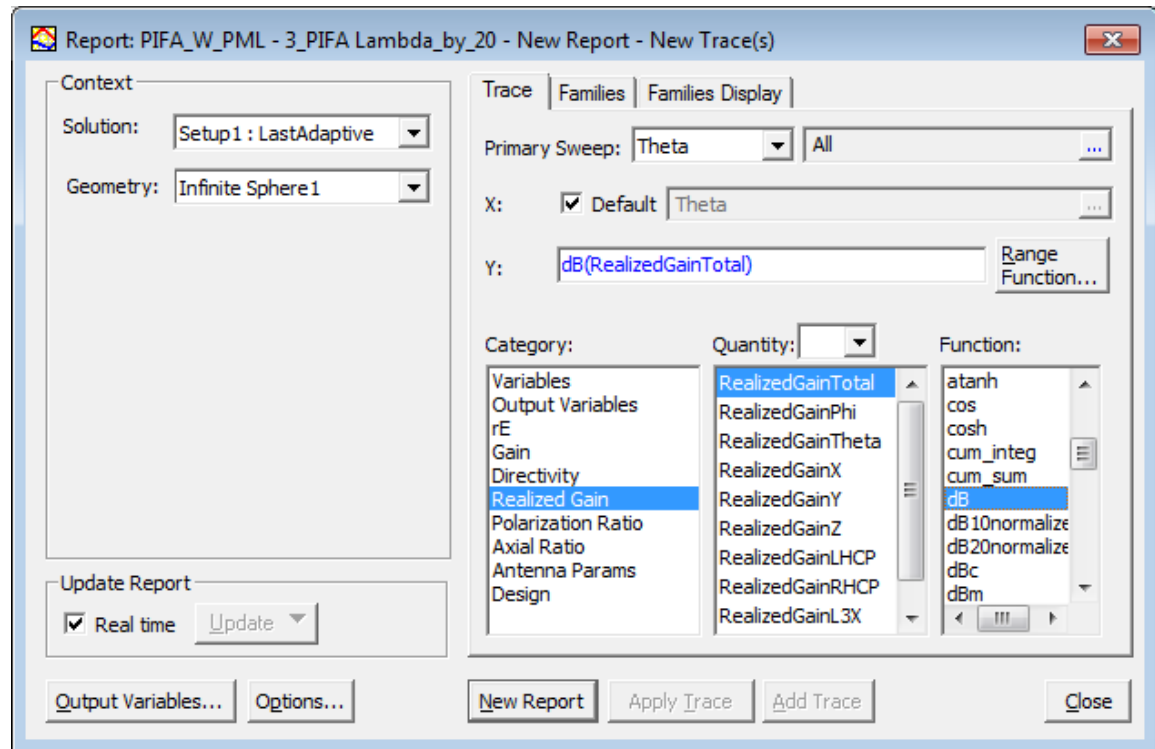
• Create Rectangular Plot

- Select the menu item **HFSS > Results > Create Terminal Solution Data Report > Rectangular Plot**
 - Solution: **Setup1: Sweep**
 - Domain: **Sweep**
 - In the **Trace** tab:
 - Category: **Terminal S Parameter**
 - Quantity: **St(1,1)**
 - Function: **dB**
 - Click the **New Report** button
 - Click the **Close** button



• Create Far Field Radiation Pattern

- Select the menu item **HFSS > Results > Create Far Fields Report > Rectangular Plot**
 - Solution: **Setup1: LastAdaptive**
 - Geometry: **Infinite Sphere1**
 - In the **Trace** tab:
 - Category: **Realized Gain**
 - Quantity: **RealizedGainTotal**
 - Function: **dB**
 - Click the **Families** Tab
 - Click the **Edit** button  associated with **Phi** variable
 - Click **0deg** in the pop-up window
 - Close the pop-up window by clicking the **X** button
 - Freq: **2.5 GHz**
 - Click the **New Report** button
 - Click the **Close** button



- **Simulation Setup**

- Select the Design “2_PIFA Lambda_by_10”
- Repeat the steps from slide 14 to 18
 - Except Minimum Radiating Distance for PML Setup: **1.5cm**
- Select the Design “3_PIFA Lambda_by_4”
- Repeat the steps from slide 14 to 18
 - Except Minimum Radiating Distance for PML Setup: **3.75cm**

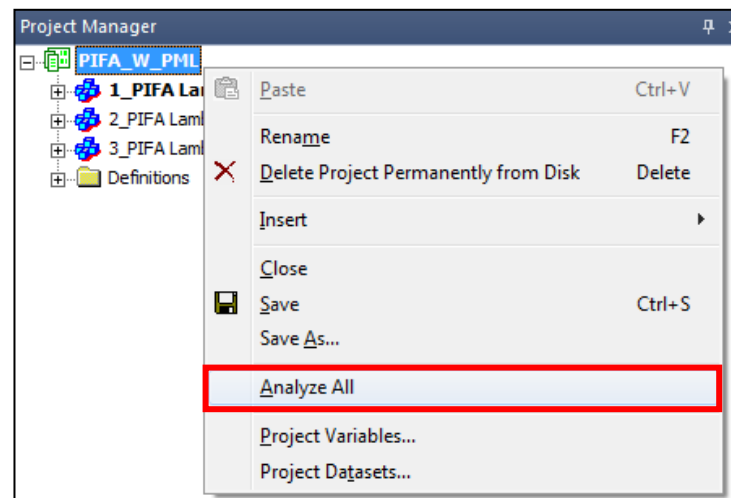
- **Save Project**

- Select the menu item **File > Save**

- **Analyze Project**

- From the **Project Manager** window, Right-click on the project **PIFA_W_PML**
 - Select **Analyze All**

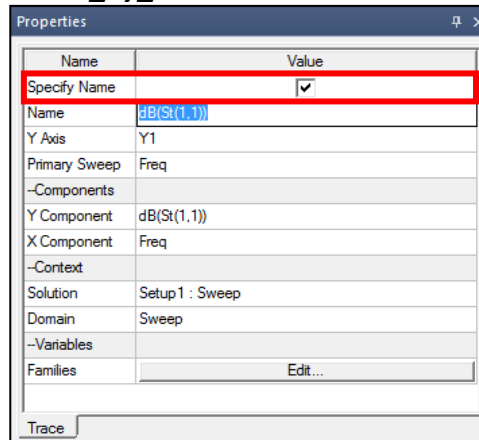
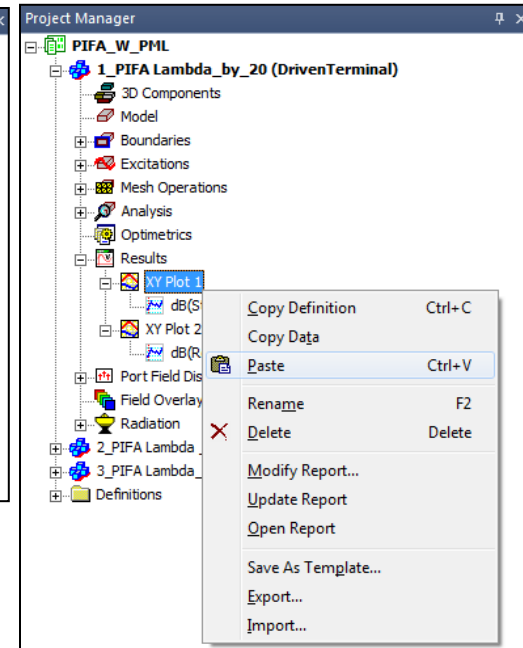
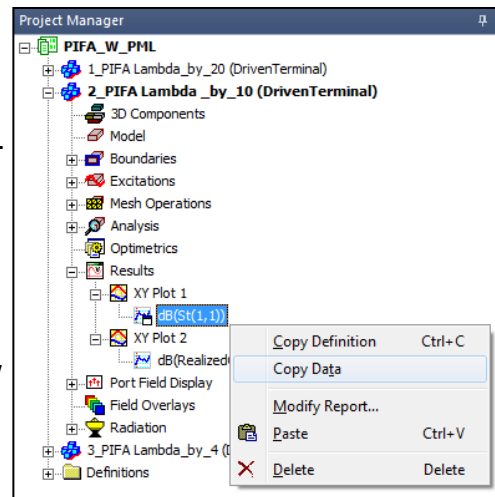
Note: This will solve all of the designs in this project.



Comparing Return Loss Results

Comparing Return Loss

- Select the Design “**2_PIFA Lambda_by_10**”
- Expand **Results**, right-click on **dB(St(1,1))** under **XY Plot 1**
 - Select **Copy Data**
- Select the Design “**1_PIFA Lambda_by_20**”
- Expand **Results**, right-click on **XY Plot 1**
 - Select **Paste**
 - Click on **dB(St(1,1))**, and from the **Properties** window
 - Check the box for **Specify Name**
 - Name: **dB(St(1,1))_Lambda_by_20**
 - Click on **dB(St(1,1))_1**, and from the **Properties** window
 - Name: **dB(St(1,1))_Lambda_by_10**



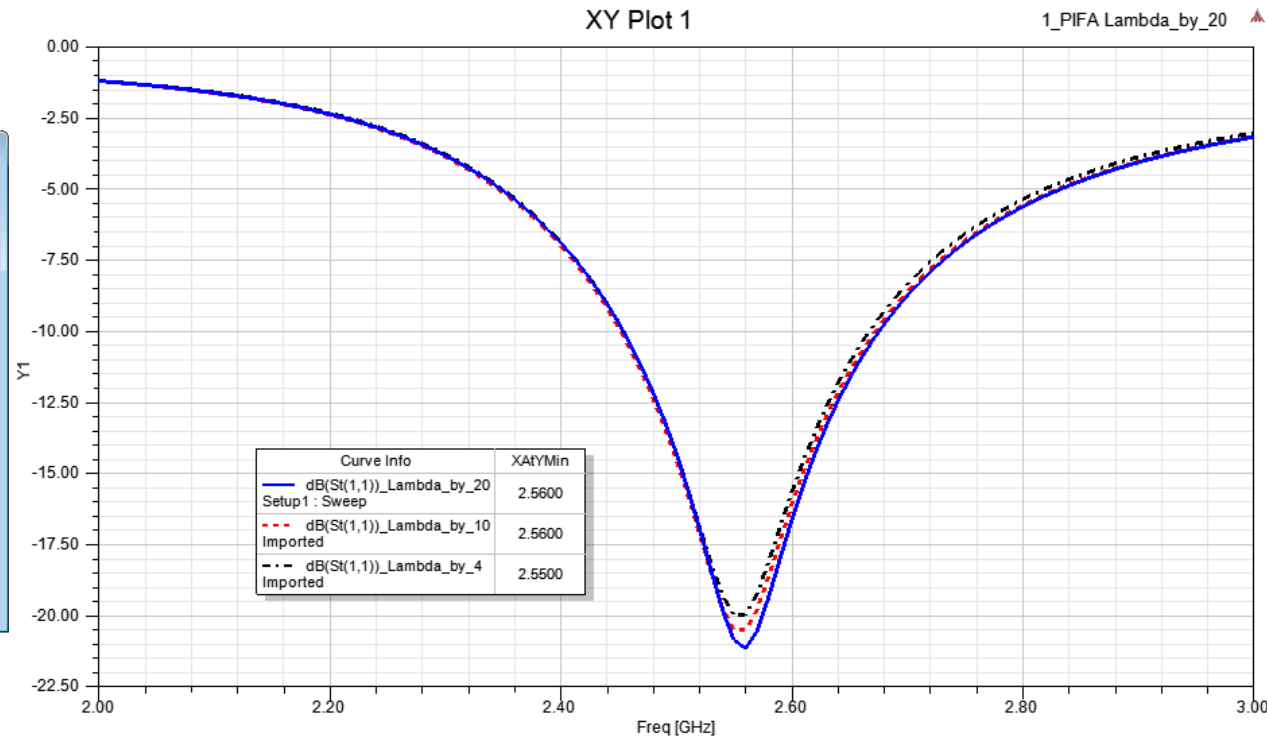
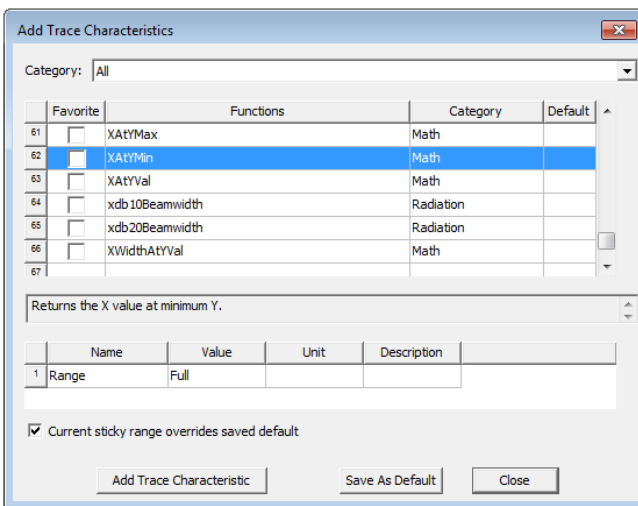
- Similarly, copy the Return Loss **dB(St(1,1))** from “**3_PIFA Lambda_by_4**” and paste into design “**1_PIFA Lambda_by_20**”
 - Rename it to **dB(St(1,1))_Lambda_by_4**

Comparing Return Loss Results

Trace characteristics

- Double-click on **XY Plot 1** in the design **1_PIFA Lambda_by_20**
- Select the menu item **Report2D > Trace Characteristics > All...**
- Select Scroll-down to the bottom and select **XatYMin**
 - Click the **Add Trace Characteristic** button
 - Click the **Close** button

Note: The results of all three are almost identical. We recommended that you use a PML with a $\lambda/10$ minimum spacing from the radiating element



Comparing Radiation Pattern

Comparing Radiation Pattern plots

- Repeat the same procedure as shown in **Slide 20** for comparing Radiation pattern between the three designs for **XY Plot 2**
- Double-click on **XY Plot 2** in the design **1_PIFA Lambda_by_20**

Trace characteristics

- Select the menu item **Report2D > Trace Characteristics > All...**
- Select Scroll-down to the bottom and select **YatXVal**
 - X Value: **90deg**
 - Click the **Add Trace Characteristic** button
 - Click the **Close** button

Add Trace Characteristics

Category: All

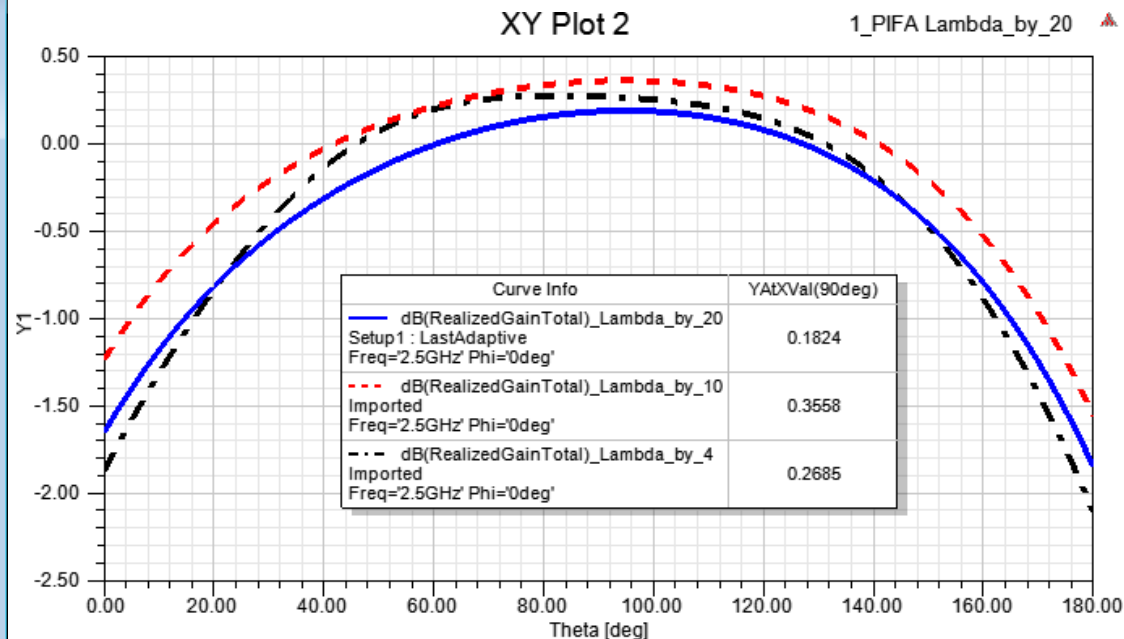
	Favorite	Functions	Category	Default
64	☐	xdb10Beamwidth	Radiation	
65	☐	xdb20Beamwidth	Radiation	
66	☐	XWidthAtYVal	Math	
67	☐	YAtXMax	Math	
68	☐	YAtXMin	Math	
69	☒	YAtXVal	Math	

Returns the Y value at the first occurrence of X value.

	Name	Value	Unit	Description
1	X Value	90	deg	X value at which we need to find Y
2	Range	Full		

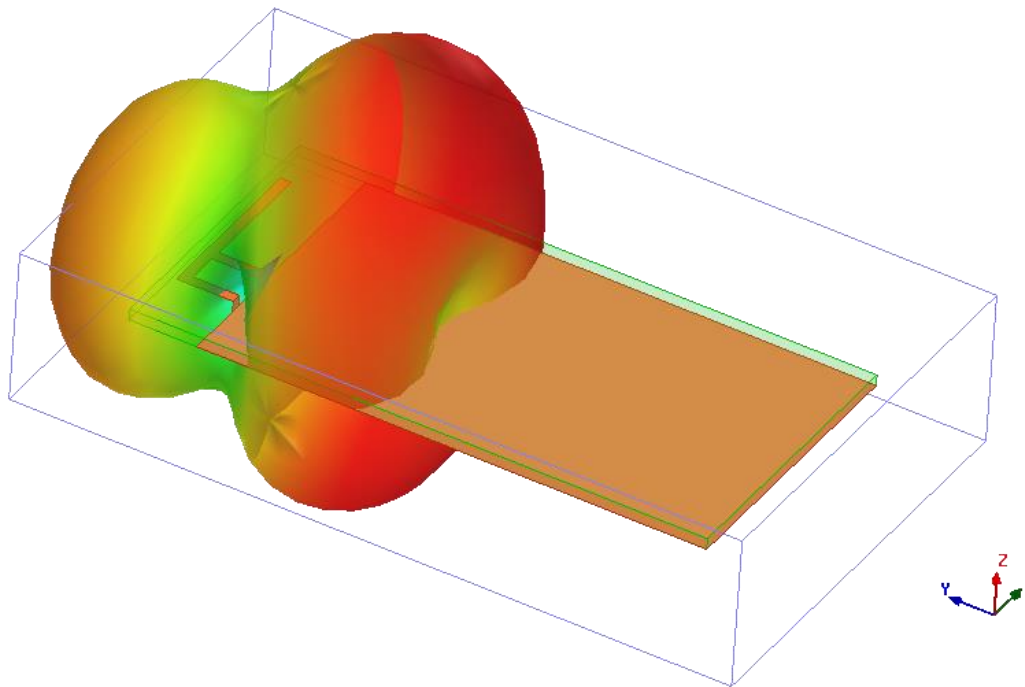
☒ Current sticky range overrides saved default

Add Trace Characteristic Save As Default Close



- **Opening a Project**

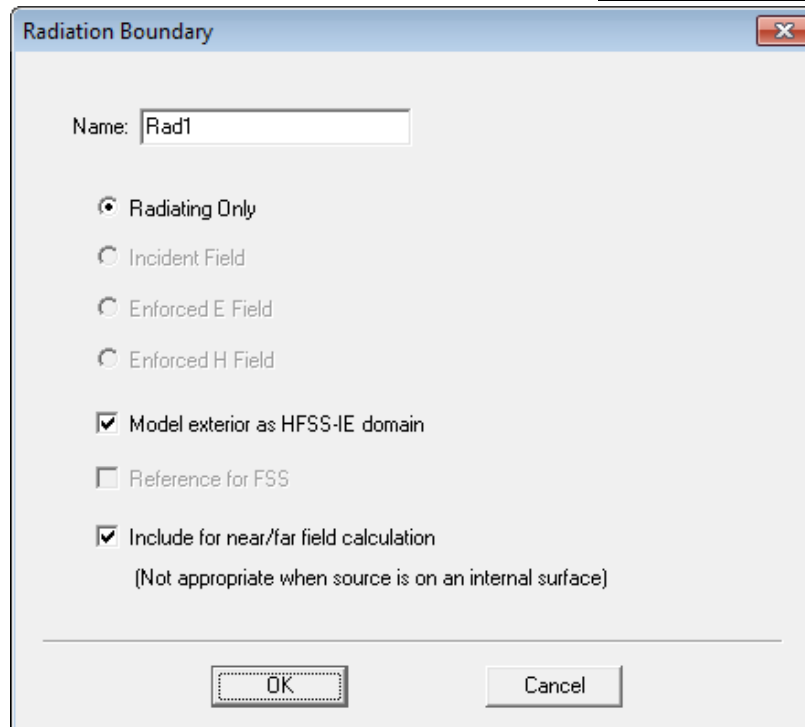
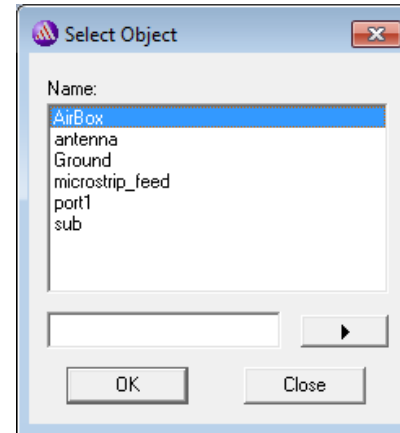
- In the ANSYS Electronics Desktop, select the menu item **File > Open**
 - Browse to the folder containing the file **PIFA_Training.aedt** and select **Open**
 - Select the menu item **File > Save As**.
 - From the **Save As** window, type the Filename: **PIFA_W_FEBI.aedt**



Radiation Boundary

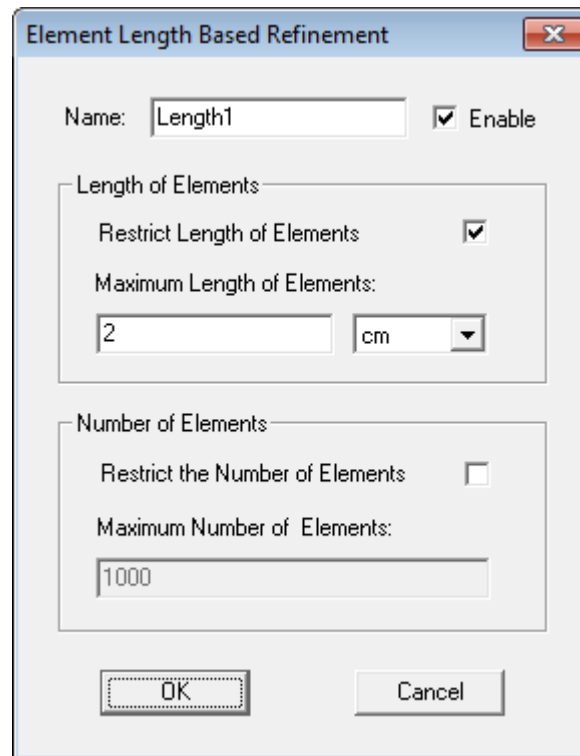
• Applying Radiation Boundary

- Select the menu item **Edit > Select > Objects**
- Select the menu item **Edit > Select > By Name...**
 - Select **AirBox** and click **OK**
- Select the menu item **Edit > Select > All Object Faces**
- Select the menu item **HFSS > Boundaries > Assign > Radiation...**
- Model exterior as HFSS-IE domain: ☒ **Checked**
 - Click the **OK** button



- **Applying Mesh Operation on Radiation Boundary**

- Select the menu item **Edit > Select > Objects**
- Select the menu item **Edit > Select > By Name...**
 - Select **AirBox** and click **OK**
- Select the menu item **Edit > Select > All Object Faces**
- Select the menu item **HFSS > Mesh Operations > Assign > On Selection > Length Based**
 - Maximum Length of Elements: **2cm**
- Click the **OK** button

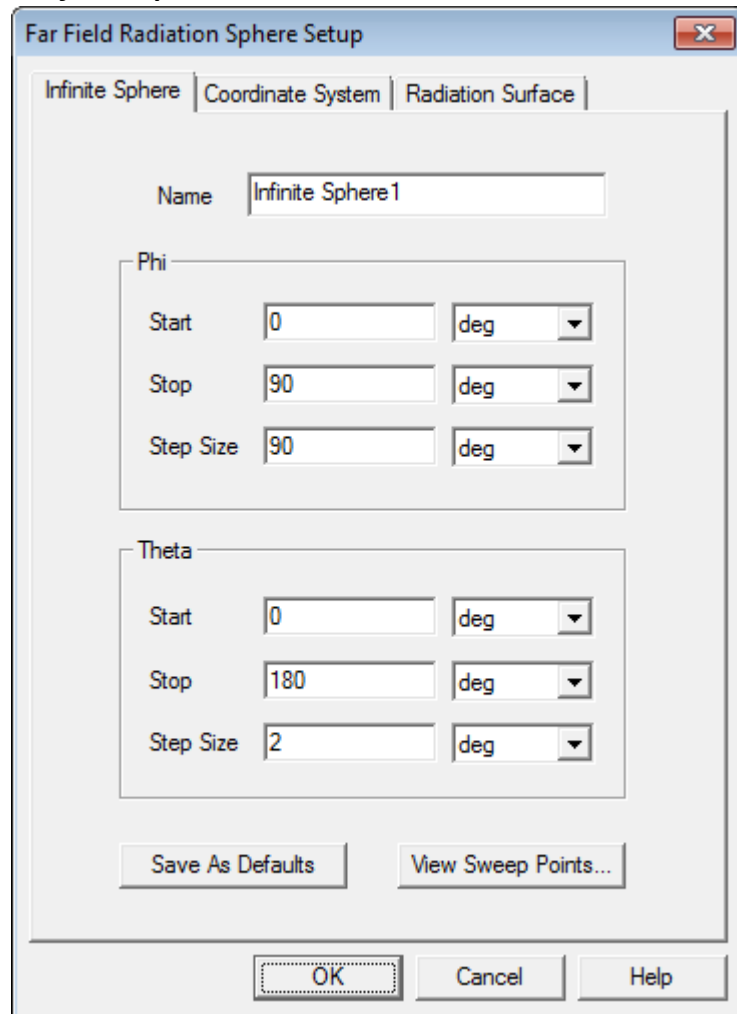


- **Create a Radiation Setup**

- Select the menu item **HFSS > Radiation > Insert Far Field Setup > Infinite Sphere...**
 - In the **Infinite Sphere** Tab
 - Name: **Infinite Sphere 1**
 - Phi:
 - Start: **0 deg**
 - Stop: **90 deg**
 - Step Size: **90 deg**
 - Theta:
 - Start: **0 deg**
 - Stop: **180 deg**
 - Step Size: **2 deg**
 - Click the **OK** button

- **Model Validation**

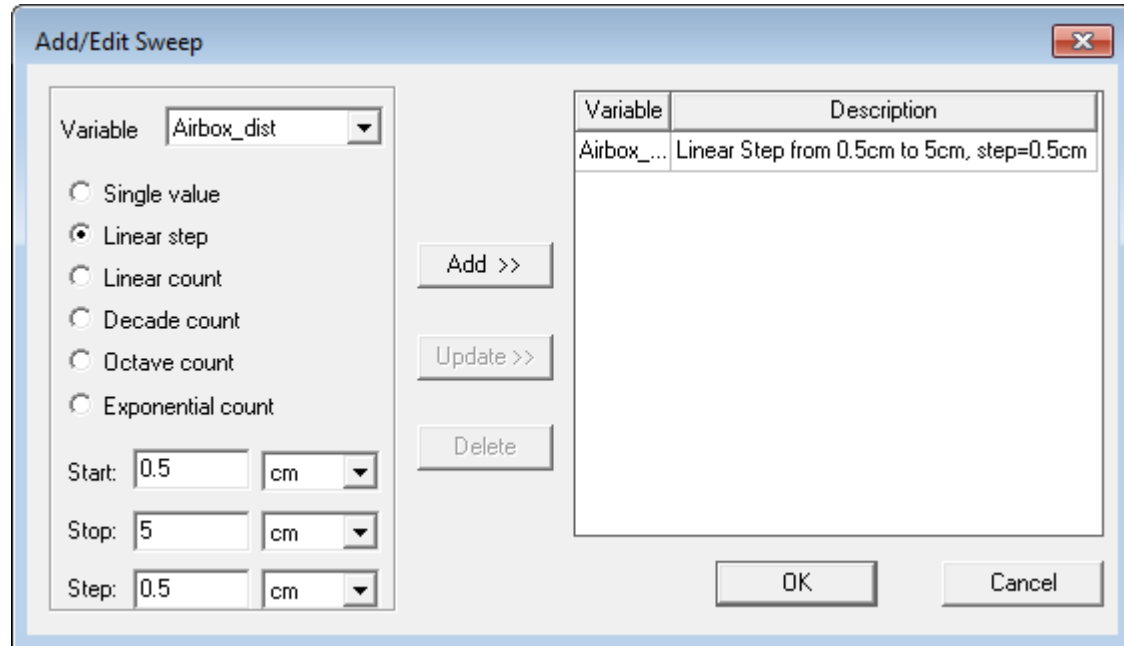
- Select the menu item **HFSS > Validation Check**
- Click the **Close** button



- **Creating a Parametric Sweep**

- Select the menu item **HFSS > Optimetrics Analysis > Add Parametric...**

- In the **Sweep Definitions** tab.
 - Click the **Add...** button
 - Variable: **AirBox_dist**
 - Select the radio button: **Linear step**
 - Start: **0.5 cm**
 - Stop: **5 cm**
 - Step: **0.5 cm**
 - Click the **Add>>** button
 - Click the **OK** button
 - Click the **Options** tab.
 - Save Fields and Mesh: ☒ **Checked**
 - Click the **OK** button



- **Save Project**

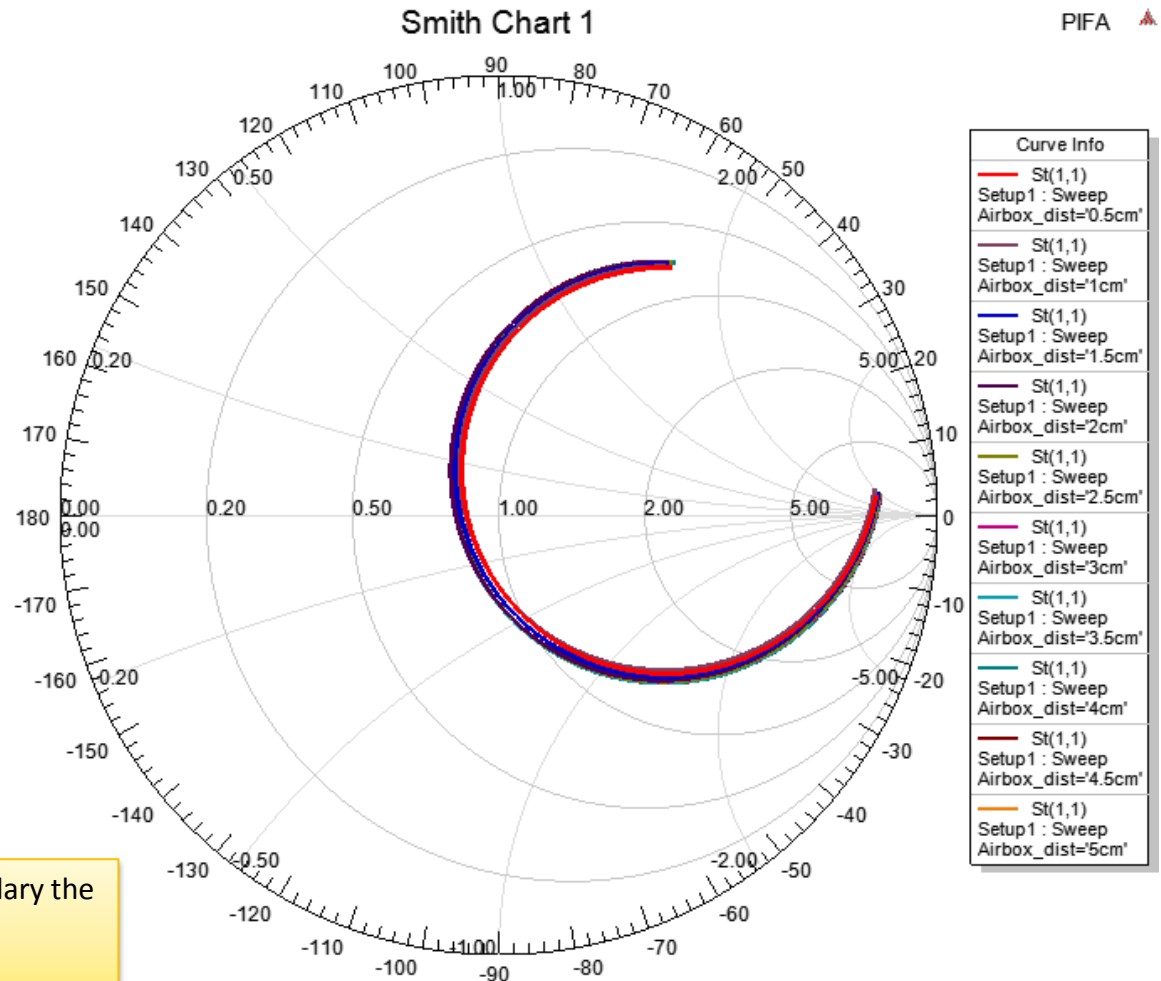
- Select the menu item **File > Save**

- **Analyze the Project**

- Select the menu item **HFSS > Analyze All**

- **Create Smith Chart**

- Select the menu item **HFSS > Results > Create Terminal Solution Data Report > Smith Chart**
 - Solution: **Setup1: Sweep**
 - In the **Trace** tab:
 - Category: **Terminal S Parameter**
 - Quantity: **St(1,1)**
 - Function: **<none>**
 - Click the **New Report** button
 - Click the **Close** button



Note: With FE-BI enabled on the radiation boundary the air box volume does not effect the return loss measurement an appreciable amount

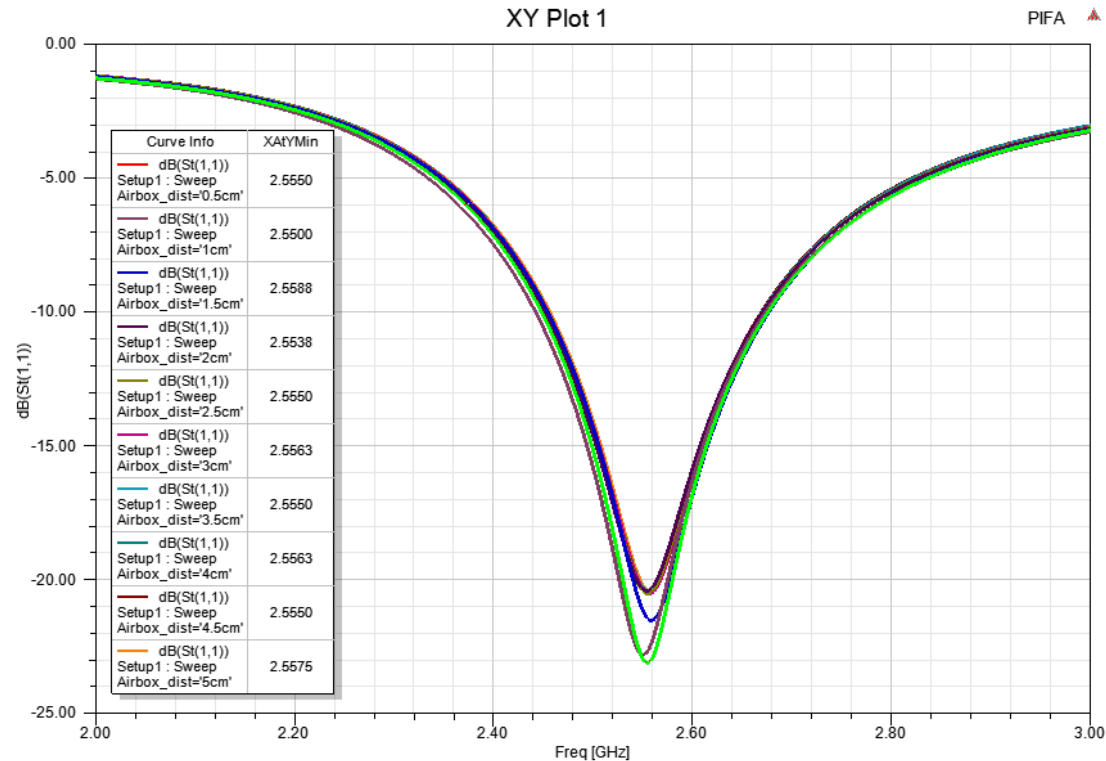
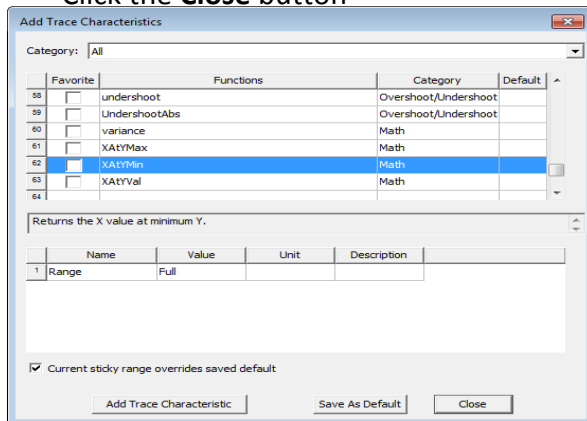
Rectangular Plot

• Create Rectangular Plot

- Select the menu item **HFSS > Results > Create Terminal Solution Data Report > Rectangular Plot**
 - Solution: **Setup1: Sweep**
 - Domain: **Sweep**
 - In the **Trace** tab:
 - Category: **Terminal S Parameter**
 - Quantity: **St(1,1)**
 - Function: **dB**
 - Click the **New Report** button
 - Click the **Close** button

• Trace characteristics

- Right-click in XY Plot1 (in Report window)
- Select **Trace Characteristics > All...**
 - Scroll-down to the bottom and select **XatYMin**
 - Click the **Add Trace Characteristic** button
 - Click the **Close** button

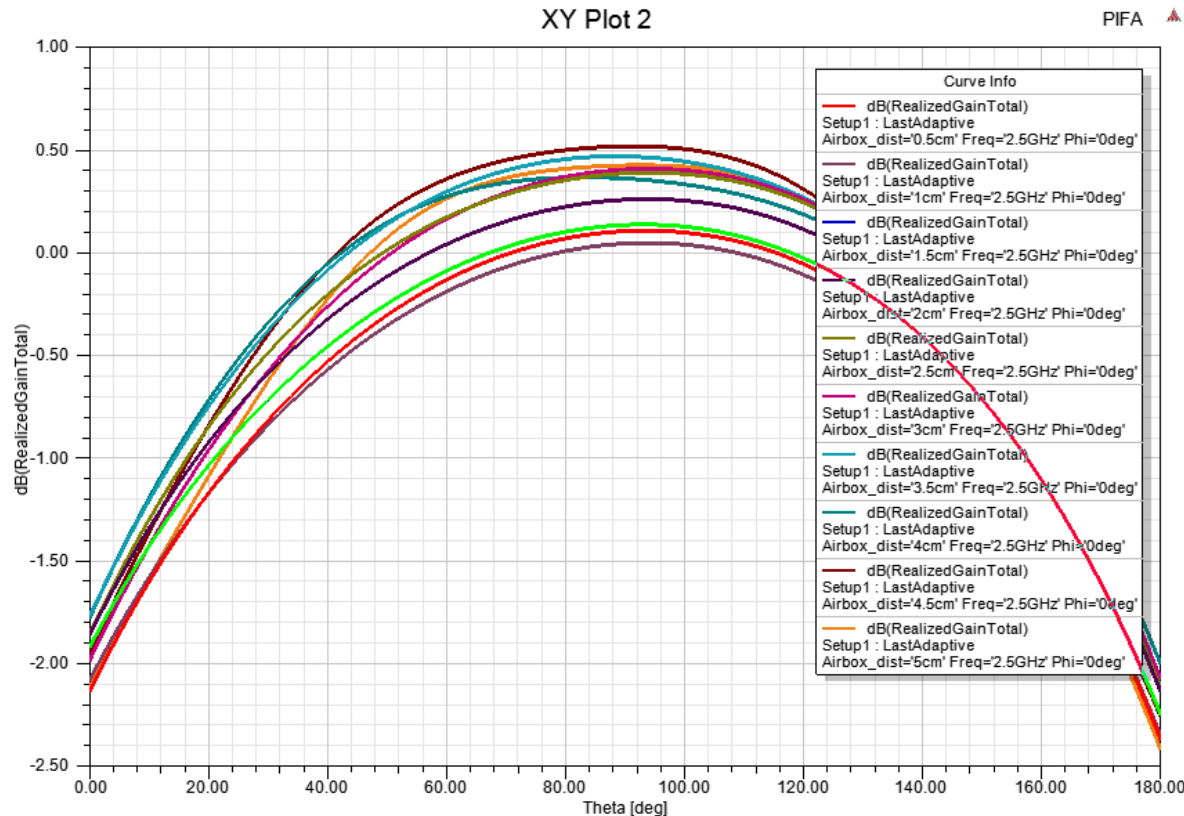


Note: The variation in the resonant frequency is +/- .2%, FE-BI gives accuracy at any air volume size

• Create Far Field Radiation Pattern

- Select the menu item **HFSS > Results > Create Far Fields Report > Rectangular Plot**

- Solution: **Setup1: LastAdaptive**
- Geometry: **Inifinite Sphere1**
- In the **Trace** tab:
 - Category: **Realized Gain**
 - Quantity: **RealizedGainTotal**
 - Function: **dB**
- Click the **Families** Tab
 - Click the **Edit** button  associated with **Phi** variable
 - Click **0deg** in the pop-up window
 - Close the pop-up window by clicking the **X** button
 - Freq: **2.5 GHz**
 - Airbox_dist: **All**
- Click the **New Report** button
- Click the **Close** button



Note: The variation in the gain is 0.1dB, FE-BI gives accuracy at any air volume size